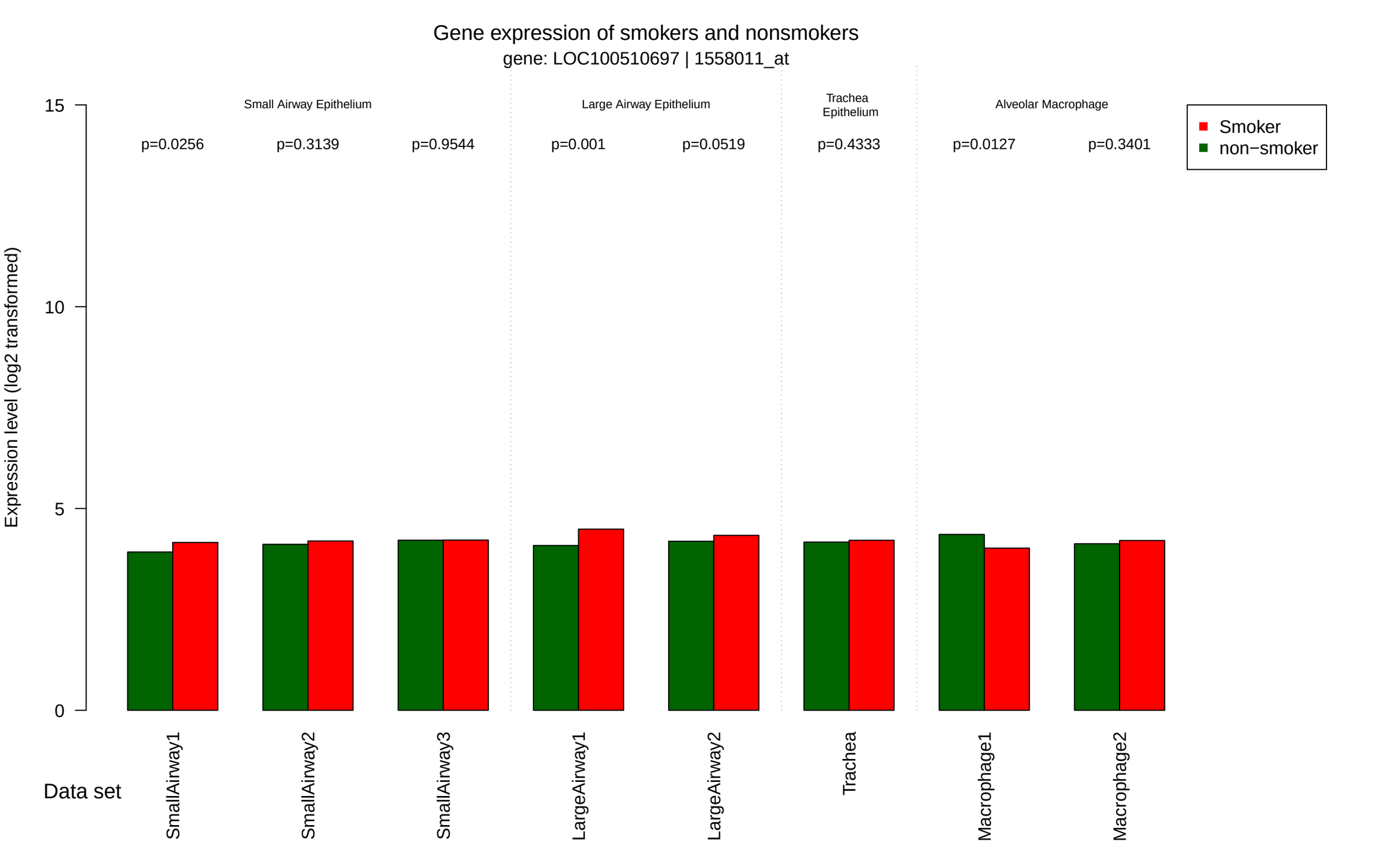
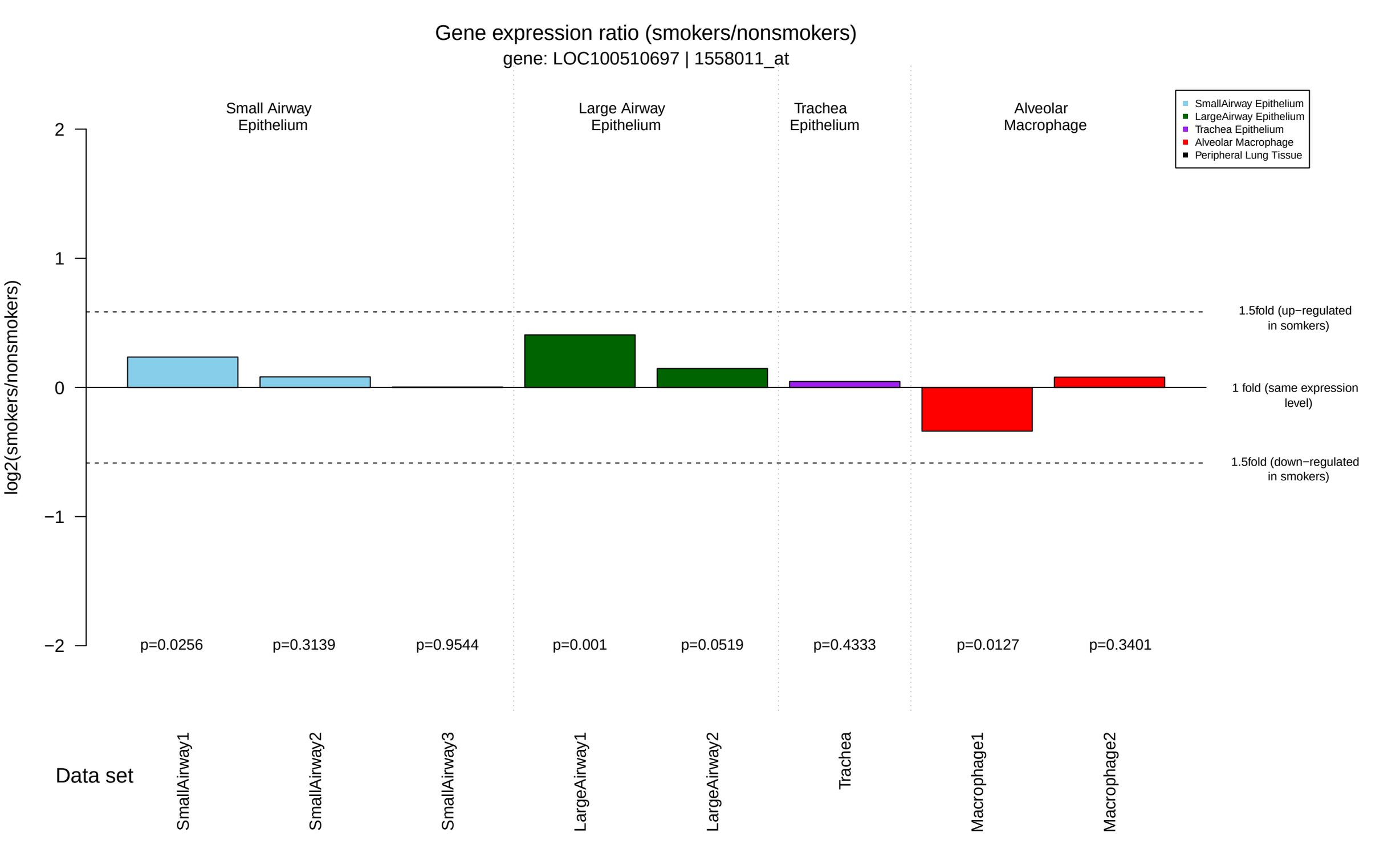
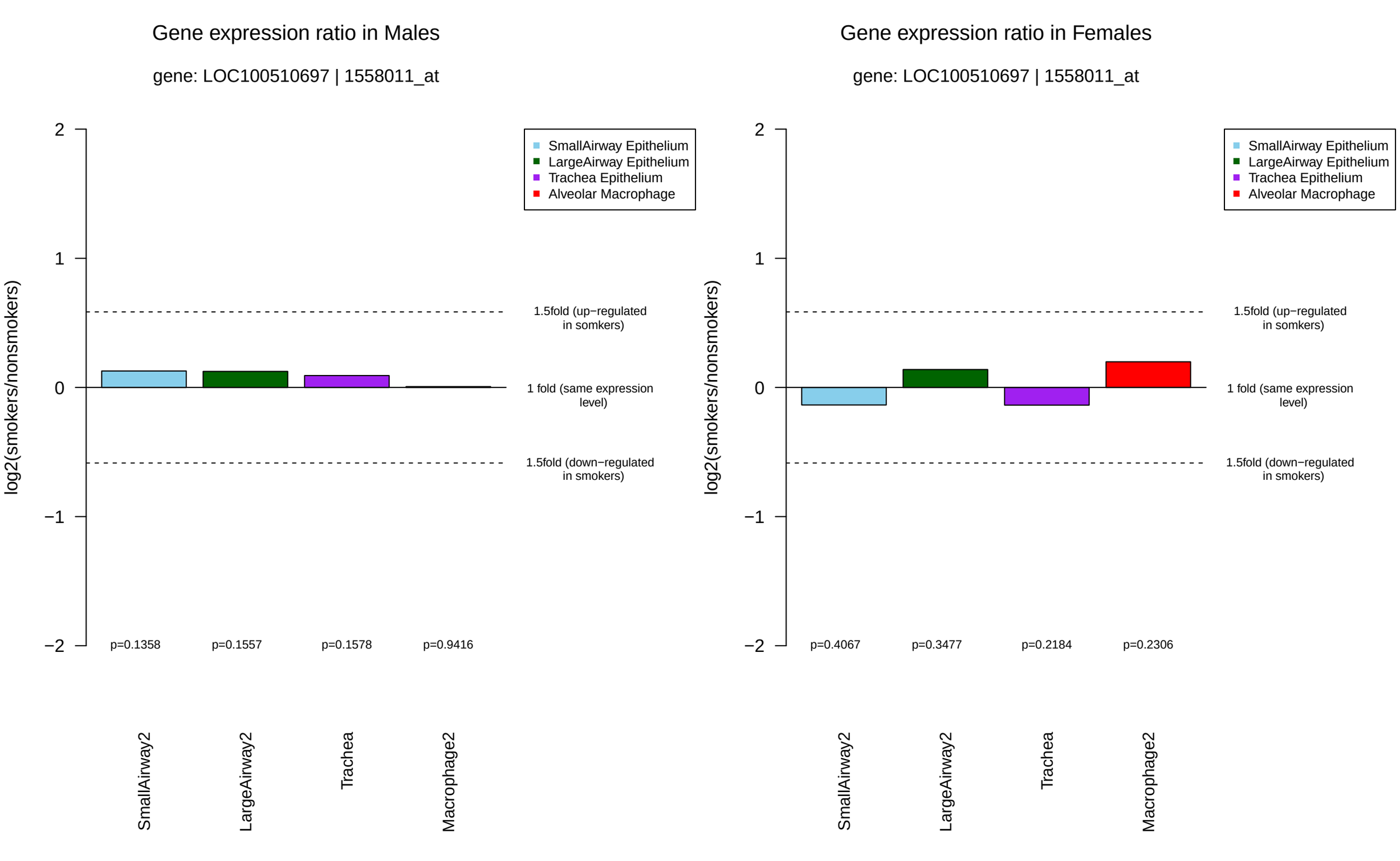
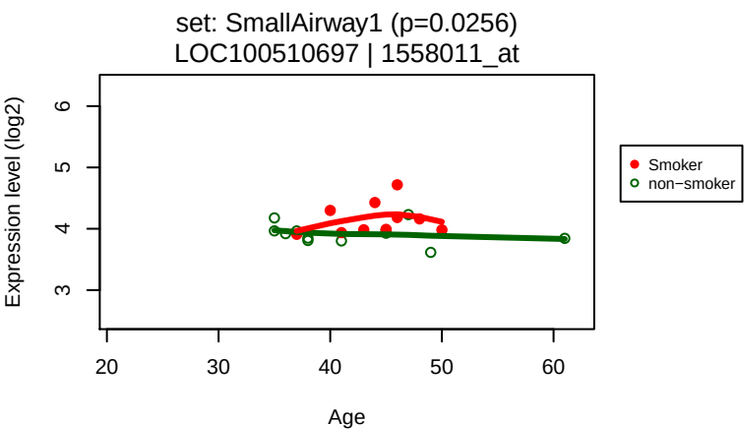
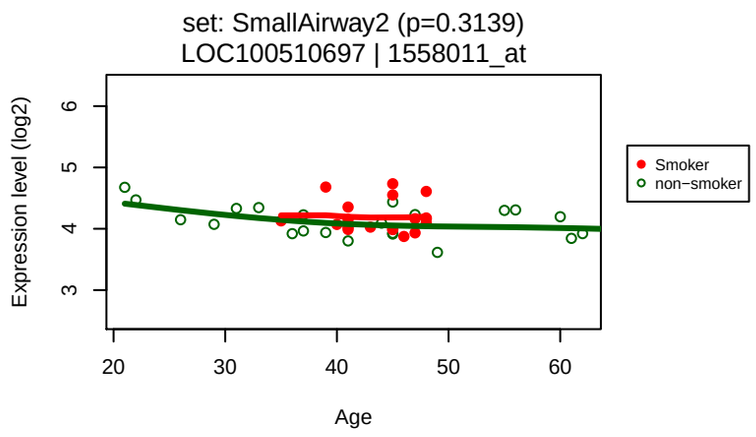
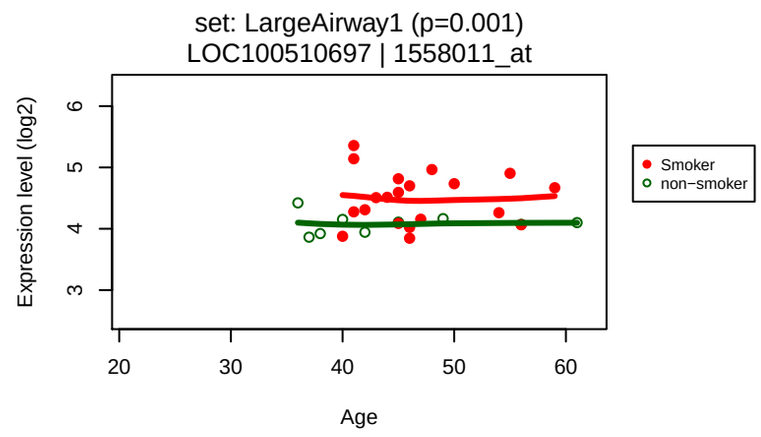

Gene expression of smokers and nonsmokers
gene: LOC100510697 | 1558011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0256
p=0.3139
p=0.9544
p=0.001
p=0.0519
p=0.4333
p=0.0127
p=0.3401
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

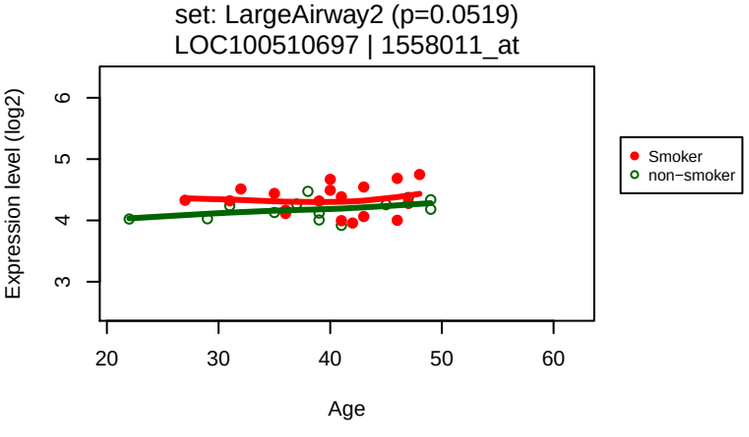

Gene expression ratio (smokers/nonsmokers)
gene: LOC100510697 | 1558011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0256
p=0.3139
p=0.9544
p=0.001
p=0.0519
p=0.4333
p=0.0127
p=0.3401
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

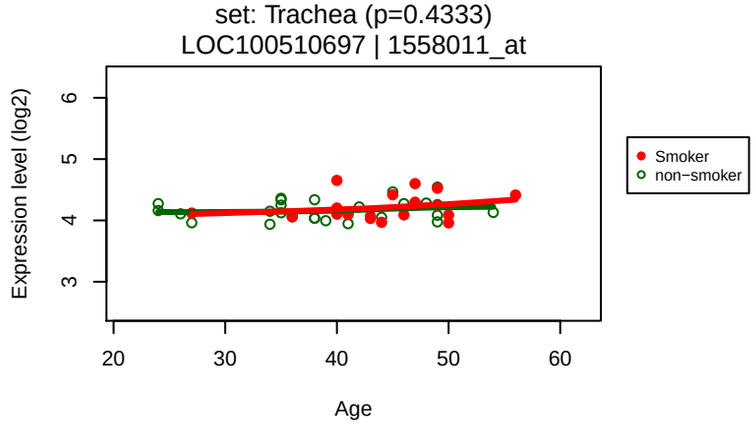

Gene expression ratio in Males
gene: LOC100510697 | 1558011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1358
p=0.1557
p=0.1578
p=0.9416
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC100510697 | 1558011_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4067
p=0.3477
p=0.2184
p=0.2306
SmallAirway2
LargeAirway2
Trachea
Macrophage2

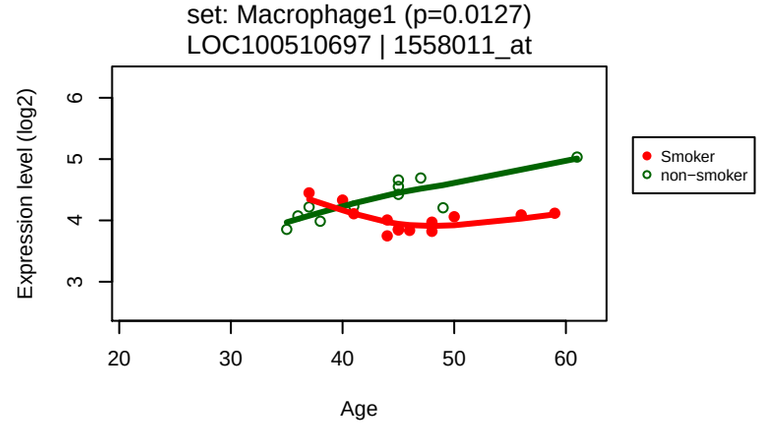

set: SmallAirway1 (p=0.0256)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker

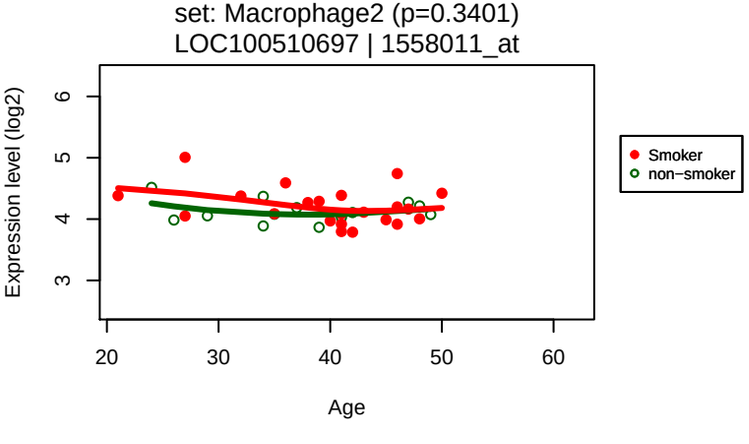

set: SmallAirway2 (p=0.3139)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker

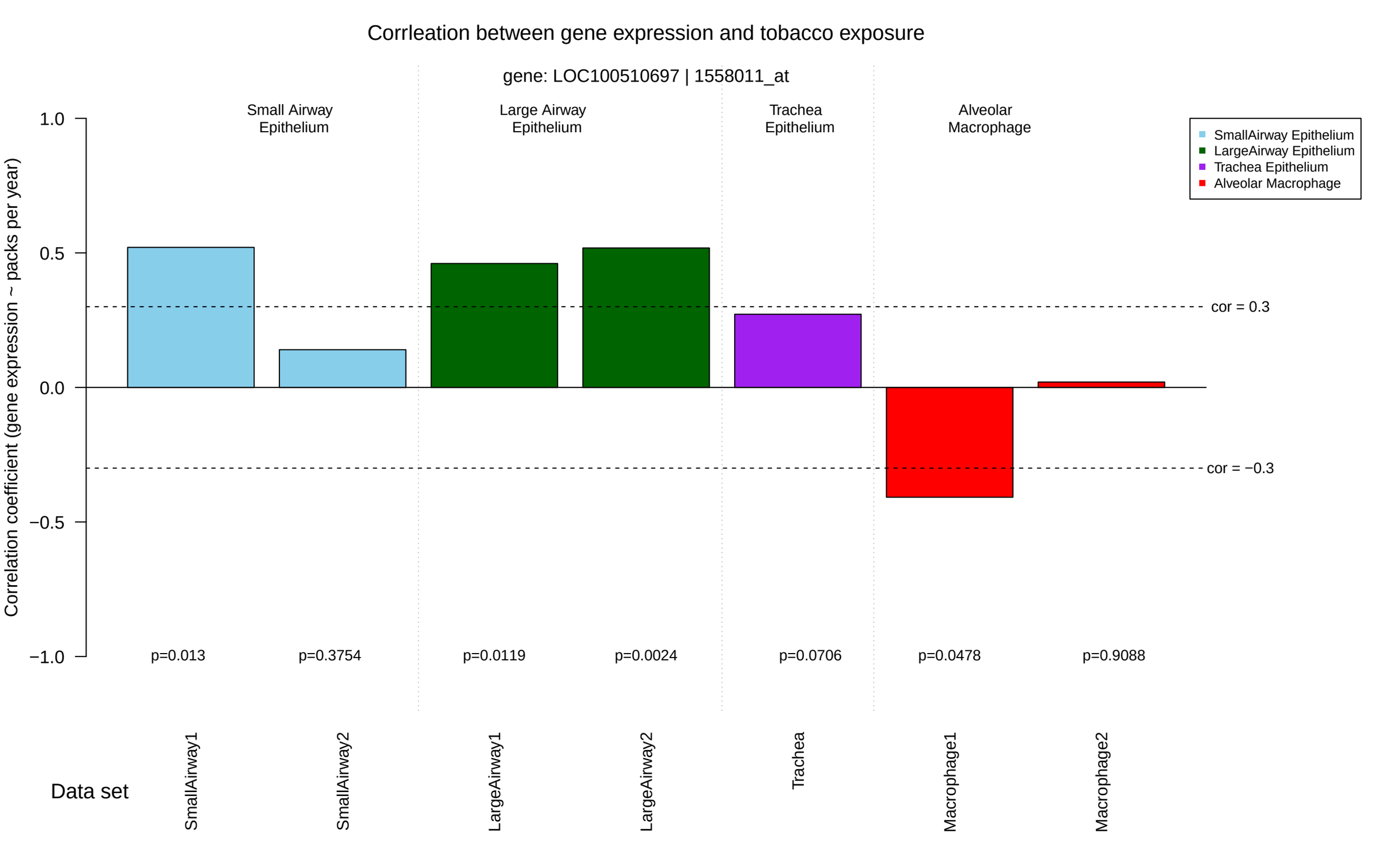

set: LargeAirway1 (p=0.001)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0519)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4333)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0127)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3401)
LOC100510697 | 1558011_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC100510697 | 1558011_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.013
p=0.3754
p=0.0119
p=0.0024
p=0.0706
p=0.0478
p=0.9088
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2